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EXPERIMENT- 4

Adder and Subtractor: Assignment

1. Verilog code and testbench for converting 8-bit binary number to gray code using data flow modeling.

Ans: Link1: https://edaplayground.com/x/7XmN

Verilog Code (Screenshot)-

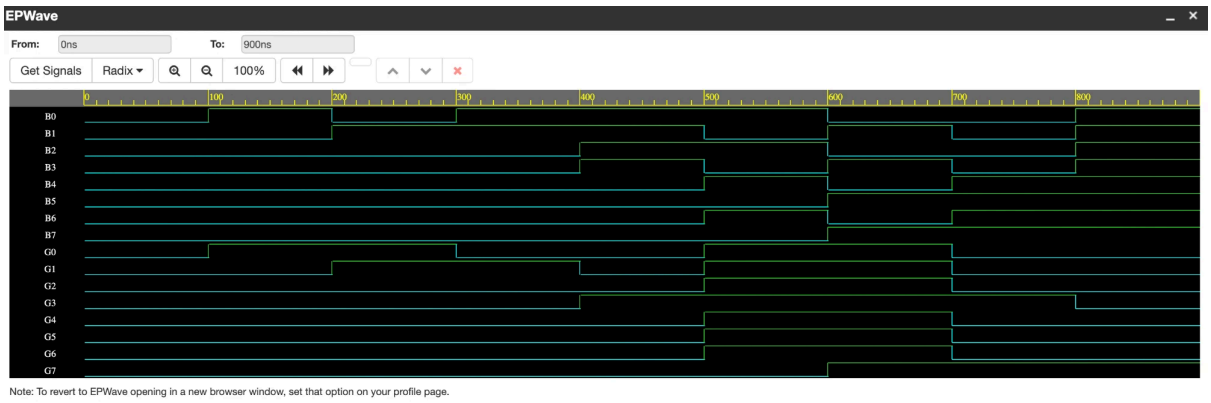
testbench.sv

```
module tb_bin_to_gray;
reg B7,B6,B5,B4,B3,B2,B1,B0;
wire G7,G6,G5,G4,G3,G2,G1,G0;
initial
begin
$dumpfile ("dump.vcd");
$dumpvars (1, tb_bin_to_gray);
#000 B7=0; B6=0; B5=0; B4=0; B3=0; B2=0; B1=0; B0=0;
#100 B7=0; B6=0; B5=0; B4=0; B3=0; B2=0; B1=0; B0=1;
#100 B7=0; B6=0; B5=0; B4=0; B3=0; B2=0; B1=1; B0=0;
#100 B7=0; B6=0; B5=0; B4=0; B3=0; B2=0; B1=1; B0=1;
#100 B7=0; B6=0; B5=0; B4=0; B3=1; B2=1; B1=1; B0=1;
#100 B7=0; B6=1; B5=0; B4=1; B3=0; B2=1; B1=0; B0=1;
#100 B7=1; B6=0; B5=1; B4=0; B3=1; B2=0; B1=1; B0=0;
#100 B7=1; B6=1; B5=1; B4=1; B3=0; B2=0; B1=0; B0=0;
#100 B7=1; B6=1; B5=1; B4=1; B3=1; B2=1; B1=1; B0=1;
#100 $stop;
end
bin_to_gray
U1(G7,G6,G5,G4,G3,G2,G1,G0,B7,B6,B5,B4,B3,B2,B1,B0);
endmodule
```

design.sv

```
module bin_to_gray (G7,G6,G5,G4,G3,G2,G1,G0,B7,B6,B5,B4,B3,B2,B1,B0);
input B7,B6,B5,B4,B3,B2,B1,B0;
output G7,G6,G5,G4,G3,G2,G1,G0;
assign G7 = B7;
assign G6 = B7^B6;
assign G5 = B6^B5;
assign G4 = B5^B4;
assign G3 = B4^B3;
assign G2 = B3^B2;
assign G1 = B2^B1;
assign G0 = B1^B0;
endmodule
```

EDA Playground waveform (Screenshot)--



2. Verilog code and testbench for full subtractor using data flow modeling.

Ans: Link2: https://edaplayground.com/x/kjj7

Verilog Code (Screenshot)-

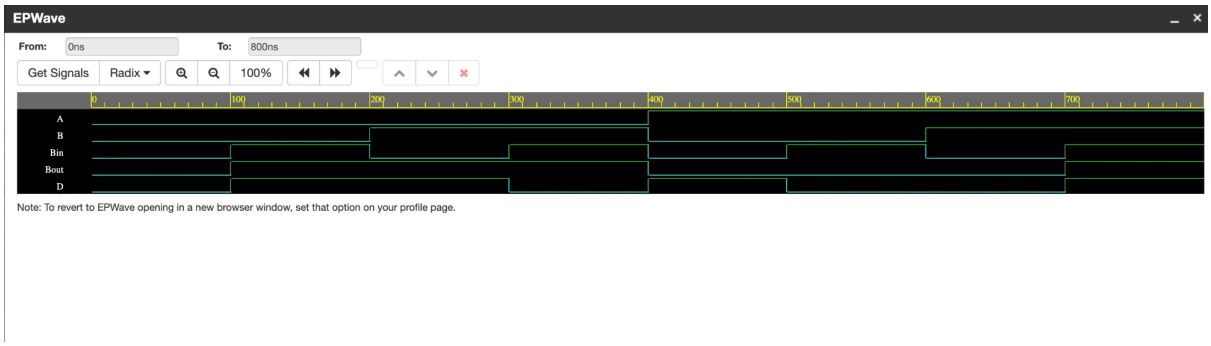
testbench.sv

```
module tb_full_sub;
reg A,B,Bin;
wire D,Bout;
initial
begin
$dumpfile ("dump.vcd");
$dumpvars (1, tb_full_sub);
#000 A=0; B=0; Bin=0;
#100 A=0; B=0; Bin=1;
#100 A=0; B=1; Bin=0;
#100 A=0; B=1; Bin=1;
#100 A=1; B=0; Bin=0;
#100 A=1; B=0; Bin=1;
#100 A=1; B=1; Bin=0;
#100 A=1; B=1; Bin=1;
#100 $stop;
end
full_subtractor U1(D,Bout,A,B,Bin);
endmodule
```

design.sv

```
module full_subtractor (D,Bout,A,B,Bin);
input A,B,Bin;
output D,Bout;
assign D = A^B^Bin;
assign Bout = (~A&B) | (~(A^B)&Bin);
endmodule
```

EDA Playground waveform (Screenshot)--



3. Verilog code and testbench for full subtractor using behavioral modeling.

Ans: Link3: https://edaplayground.com/x/DRZi

Verilog Code (Screenshot)-

testbench.sv

```
module tb_full_sub;
reg A,B,Bin;
wire D,Bout;
initial
begin
$dumpfile ("dump.vcd");
$dumpvars (1, tb_full_sub);
#000 A=0; B=0; Bin=0;
#100 A=0; B=0; Bin=1;
#100 A=0; B=1; Bin=0;
#100 A=0; B=1; Bin=1;
#100 A=1; B=0; Bin=0;
#100 A=1; B=0; Bin=1;
#100 A=1; B=1; Bin=0;
#100 A=1; B=1; Bin=1;
#100 $stop;
end
full_subtractor U1(D,Bout,A,B,Bin);
endmodule
```

design.sv

```
module full_subtractor (D,Bout,A,B,Bin);
input A,B,Bin;
output reg D,Bout;
always @(*)
begin
D = A^B^Bin;
Bout = (~A&B) | (~(A^B)&Bin);
end
endmodule
```

EDA Playground waveform (Screenshot)--